

PAPER_960: VDS Polylogarithm Li_{26} Reference

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Session: 215 **Source:** ramanujan_26d_summation.py (VDSPolylog26) **Calculator:**
VDSPolylog26Calc (CP4 #544) **CVW:** v2.0.0 compliant

Abstract

Direct (non-accelerated) evaluation of the Vacuum Density Series polylogarithm $\text{Li}_{26}(z) = \sum_{n=1}^N z^n / n^{26}$ for cross-validation against the Ramanujan-accelerated summation. Both must agree to working precision.

1. VDS Polylogarithm

$$\text{Li}_{26}(z) = \sum_{n=1}^N \frac{z^n}{n^{26}}$$

2. Cross-Validation

$$|S_{26}(z) - \text{Li}_{26}(z)| \xrightarrow{N \rightarrow \infty} 0$$

References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)